

DPL: Diverse Preference Learning Without A Reference Model

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Abstract

In direct preference alignment of LLMs, most existing methods seek to retrieve the reward function directly from preference data. However, real-world preference data often contains *diversity* in preference annotations reflective of true human preferences. Existing algorithms, including KTO (Ethayarajh et al., 2024), do not directly utilize such nuances in the annotations which limits their applicability. In this work, we propose Diverse Preference Learning (DPL), a reference model-free method that simultaneously learns a baseline desirability in LLM responses while being robust to the diversity of preference annotations. Our experiments for instruction-following on Ultrafeedback and AlpacaEval 2.0 and for text-summarization on Reddit TL;DR suggest that DPL is consistently better at learning the diversity of preferences compared to existing methods, including those that require a reference-model in memory. Apart from overall quality, we find that DPL’s completions, on average, are more honest, helpful, truthful and safe compared to existing methods.

1 Introduction

Well-known approaches to human preference learning in LLMs, including RLHF-based methods like Proximal Policy Optimization (PPO; Schulman et al. (2017)), require costly reward models to guide alignment, limiting their utility in practical resource-constrained settings. In contrast, offline algorithms like Direct Preference Optimization (DPO; Rafailov et al. (2024)) provide better compute-efficiency by optimizing directly on preference data *without* an explicit reward model. Reference-model free variants like ORPO (Hong et al., 2024) and SIMPO (Meng et al., 2024) do this without a reference model in memory. At the core of direct alignment methods is the Bradley-Terry (BT) model (Bradley and Terry, 1952), which

models preferences as implicit reward differences between “good” and “bad” responses.

While appealing for their simplicity, the BT model assumption that true human preferences directly correlate with model-assigned implicit reward differences is problematic: firstly, popular preference datasets like Ultrafeedback (Cui et al., 2024) contain a non-negligible amount ($\sim 17\%$) of weak preference pairs that are rated “equally” (reward difference ~ 0) by high-capacity automatic evaluators like GPT-4. When the preference signal is weak, common direct alignment frameworks can lead to significant policy degeneracy or underfitting, especially under finite data settings (Azar et al., 2024). Additionally, such underfitting can effectively limit such methods from learning a “baseline desirability” of all available responses, especially for weak preference pairs where both examples are rated low. (See Figure 1 for an illustration.)

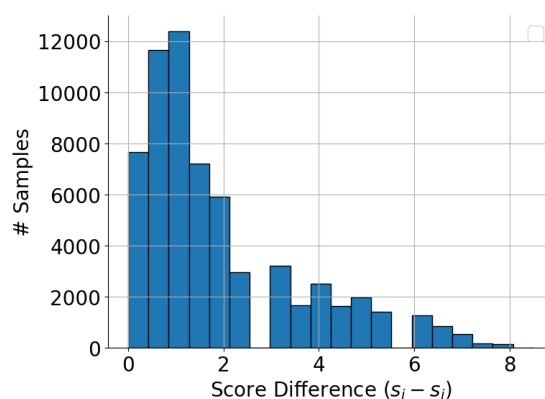


Figure 1: Reward estimates (preference strength) from GPT-4 on the Ultrafeedback dataset (Cui et al., 2024). Diverse Preference Learning (DPL) learns both a “baseline desirability” as well as the “relative goodness” between responses, outperforming competitive baselines including KTO (Ethayarajh et al., 2024) on popular preference learning benchmarks (see Table 2).

In particular, reward underspecification in the Bradley-Terry model (Bertrand et al., 2023) can hinder these frameworks from capturing the nu-

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anced preferences or relative “goodness” between responses—which is arguably available in widely-used datasets. This is true when the preference distribution¹ is varied or more nuanced, i.e., the reward differences fall within a *spectrum* as shown in Figure 1. Even more sophisticated methods like Kahneman-Tversky Optimization (KTO; [Ethayarajh et al. \(2024\)](#)), often reduce pointwise preferences to binary labels, sacrificing the granularity needed to fully leverage rich annotations.

In this work, we introduce **Diverse Preference Learning (DPL)**, a novel reference-model-free alignment framework that enables policies to capture the rich, often overlooked nuances of human feedback. DPL simultaneously learns baseline desirability and optimizes for the relative strength of preference labels, resulting in a more accurate and nuanced representation of true human preferences.

Our novel contributions are:

- We provide a practical implementation of **Diverse Preference Learning (DPL)**, addressing limitations in BT model-based approaches with two simple reward formulations derived from popular frameworks like ORPO ([Hong et al., 2024](#)) and SIMPO ([Meng et al., 2024](#)), consistent with a reference model-free setting.
- We demonstrate the effectiveness of DPL in learning diverse preferences on two popular benchmarks: Ultrafeedback ([Cui et al., 2024](#)) and Reddit TL;DR (CNN/Daily Mail) ([Völske et al., 2017](#); [Stiennon et al., 2020](#)), using both LLM-as-a-judge based evaluation as well as a Reward Model.
- Using [Cui et al. \(2024\)](#)’s annotation framework, we provide a “first-past-the-post” (FPP)-style evaluation of DPL on multidimensional preference axes such as *honesty*, *helpfulness*, *truthfulness* and *safety*, where it significantly outperforms competitive baselines such as Supervised Finetuned (SFT) models, ORPO, SIMPO, and KTO. Additionally, DPL also outperforms such baselines on popular leaderboards such as AlpacaEval 2.0.

2 Related Works

Reinforcement Learning from Human Feedback (RLHF) aims to harmonize large language models with human preferences and values ([Christiano](#)

¹While the true distribution of human preferences is unknown and likely multidimensional ([Tversky, 1969](#)), we assume GPT-4’s reward estimates provide a reasonable approximation, particularly when direct annotations are unavailable.

[et al., 2017](#); [Ziegler et al., 2019](#); [Ouyang et al., 2022](#); [Bai et al., 2022](#)). The conventional RLHF pipeline typically consists of three primary phases: supervised fine-tuning ([Zhou et al., 2023](#); [Taori et al., 2023](#); [Geng et al., 2023](#); [Conover et al., 2023](#); [Köpf et al., 2023](#); [Ding et al., 2023](#); [Wang et al., 2024b](#); [Chen et al., 2024a](#); [Xia et al., 2024](#)), reward model training ([Gao et al., 2023](#); [Luo et al., 2023](#); [Chen et al., 2024b](#); [Lightman et al., 2023](#); [Havrilla et al., 2024](#); [Lambert et al., 2024](#)), and policy optimization ([Schulman et al., 2017](#); [Anthony et al., 2017](#)). More recently, supervised offline contrastive learning approaches like DPO ([Rafailov et al., 2024](#)), IPO ([Azar et al., 2024](#)), e-DPO ([Fisch et al., 2024](#)), KTO ([Ethayarajh et al., 2024](#)), and DPO-Positive ([Pal et al., 2024](#)) have been proposed. These approaches include a reference model in memory as a baseline policy and intend to minimize compute by avoiding online sampling for reward maximization. Specifically, algorithms like DPO and KTO seek to retrieve optimal policies that analytically derive from the KL-constrained optimization problem ([Ziebart et al., 2008](#)); DPO seeks to align with human preferences, and KTO, with human preferences and utility.

In contrast, reference-free approaches like CPO ([Xu et al., 2024](#)), ORPO ([Hong et al., 2024](#)) and SIMPO ([Meng et al., 2024](#)) often assume the reference model is a uniform distribution, and proposes alternative Bradley-Terry reward formulations, which increase efficiency by dispensing with the need for an additional reference model. [Dubey et al. \(2024\)](#), [Dong et al. \(2024\)](#) and [Hong et al. \(2024\)](#) have explored including an SFT-based regularization on preferred responses directly in the alignment objective, whereas [Yang et al. \(2024\)](#) add an SFT term for more generalized reward learning and [Nath et al. \(2024\)](#) simultaneously learn preferences and rewards with reward-distillation. Specifically, our work can be seen extending this line of work but with a novel unlikelihood ([Welleck et al., 2020](#)) penalty on dispreferred responses, that along with our nuanced preference categorization, allows more diverse preference alignment.

3 Background: Implicit Rewards (r^*)

In “supervised” preference alignment, we are given a dataset of pairwise preference data $\mathcal{D}_{\text{pref}} = \{x, y_w, y_l, s_{ij}\}_{i=1}^N$, where x denotes the prompt or context, y_w and y_l are the preferred and dispreferred completions respectively, and s_{ij} represents the strength of the relative preference between y_w

and y_l .² The parameters of the aligned policy π_θ are then estimated “directly” on $\mathcal{D}_{\text{pref}}$ by formulating the preference of y_w over y_l using the Bradley Terry (BT) model as:

$$p^*(y_w \succ y_l | x) = \sigma(r^*(x, y_w) - r^*(x, y_l)) \quad (1)$$

where $p^*(y_w \succ y_l | x)$ denotes the true probability that humans prefer y_w over y_l , $\sigma(\cdot)$ is the sigmoid function and $r^*(x, y)$ is a latent reward function estimated over the observed preference data.

Reference model-based implicit rewards The core idea in RLHF is that π_θ should not diverge too much from the reference model π_{ref} , the latter being the initialization point of π_θ in most direct alignment algorithms like DPO (Rafailov et al., 2024), after undergoing supervised finetuning (SFT). In DPO, this divergence constraint between π_θ and π_{ref} is realized by choosing a very specific formulation of $r^*(x, y)$, which is an analytical derivative of the KL-constrained optimal policy π_θ^* , defined as $\frac{1}{Z(x)} \pi_{\text{ref}} \exp\left(\frac{1}{\beta} r^*(x, y)\right)$. On the other hand, Kahneman-Tversky Optimization (KTO; Ethayarajh et al. (2024)), optimizes π_θ using a redefined r^* where the policy learns to distinguish between y_w and a more distributional form of y_l —the idea being that humans make choices based on many alternative possibilities—unlike DPO where the comparison is only with a single y_l per sample.

Reference model-free implicit rewards In contrast, reference model-free frameworks like ORPO (Hong et al., 2024) and SIMPO (Meng et al., 2024) assume a uniform prior for $\pi_{\text{ref}} \sim \mathcal{U}(\mathcal{Y})$. These approaches, which are both computationally efficient and effective, redefine $r^*(x, y)$ in innovative ways, eliminating the need for a reference model in memory. Our approach for reference-free **Diverse Preference Learning** (DPL) draws its theoretical insights from these two reward formulations, which are given below³:

$$r_{\text{odds}}^*(x, y) = \beta \log \frac{\pi_\theta(y | x)}{1 - \pi_\theta(y | x)} \quad (2)$$

$$r_{\text{norm}}^*(x, y) = \beta \frac{\log \pi_\theta(y | x)}{|y|} \quad (3)$$

²Datasets such as Ultrafeedback (Cui et al., 2024) do not provide explicit human preference labels due to annotation costs (Lambert et al., 2024). Instead, they use Likert scale rewards from models like GPT-4 and inferred preference strengths (s_{ij}) to approximate human preferences, converting generic response pairs into y_w and y_l labels.

³For analysis, r_{odds}^* (without an SFT term) and r_{norm}^* (without a margin γ) are simplified forms of the ORPO (Hong et al., 2024) and SIMPO (Meng et al., 2024) objectives.

where β is the KL-beta parameter and $|y|$ represents the number of tokens in a response y . Intuitively, it is easy to see that even without a reference model, the complement of the likelihood in the denominator in Eq. 2 and the length-normalization factor in Eq. 3 act as “surrogates” for π_{ref} -based KL regularization. Now, applying a sigmoid function over the RHS of Eq. 2 and Eq. 3, one can then formulate the preference of y_w over y_l under the Bradley-Terry model as:

$$p^*(y_w \succ y_l | x) = \sigma(r_{\text{odds}}^*(x, y_w) - r_{\text{odds}}^*(x, y_l)) \quad (4)$$

$$p^*(y_w \succ y_l | x) = \sigma(r_{\text{norm}}^*(x, y_w) - r_{\text{norm}}^*(x, y_l)) \quad (5)$$

The implication here is that the BT model allows us to represent the true preference distribution (p^*) in terms of the latent reward functions. As such, a naive way to estimate the parameters of these two reward formulations, and thereby retrieve the aligned π_θ , is to maximize the log-likelihood of RHS of Eq. 4 and Eq. 5. However, if the true preferences are deterministic, i.e., if $p^*(y_w \succ y_l) \in \{0, 1\}$, or $s_{ij} \gg c$ (surpassing a baseline preference strength c), π_θ can overfit to BT-assigned reward estimates and underfit the true preferences, especially since the number of parameters in π_θ typically exceeds the number of (y_w, y_l) pairs in the preference training dataset, as argued previously in Azar et al. (2024).

For clarity, let us examine a scenario where two responses, y_w and y_l , exist such that the preference probability $p^*(y_w \succ y_l) = 1$, i.e., y_w is consistently favored over y_l . The Bradley-Terry model would then require that the difference ($r_{\text{odds}}^*(x, y_w) - r_{\text{odds}}^*(x, y_l)$) must approach $+\infty$ to comply with Eq. 4. Substituting Eq. 2 in Eq. 4, one can clearly see that the log-probability ratio is $\log\left(\frac{\pi_\theta(y_w)}{\pi_\theta(y_l)}\right) \rightarrow \infty$, which implies $\pi_\theta(y_l) \rightarrow 0$, especially *without* a reference model to control for this ratio, irrespective of the *any* finite KL-regularization parameter β . Such a scenario leads to π_θ being underfitted to the preference data in these reward function parameter estimations. Additionally, if $p^*(y_w \succ y_l) < 1$ for a pair which is possible when relative preference strengths lie on a spectrum, π_θ can *empirically* estimate this probability as 1 under finite preference data. This can lead to the policy empirically overfitting such BT-assigned infinite rewards and create substantial degeneracies where the aligned π_θ emits tokens that are not even present in the training data, especially

under finite-data settings (Fisch et al., 2024). See Appendix A for an equivalent example for r_{norm}^* .

4 Diverse Preference Learning Framework (DPL)

To prevent such degeneracies, we aim to define the DPL objective such that it (a) is reference model-free for computational efficiency, (b) uses a BT model-based loss to learn from pairwise contrastive preference data, (c) regularizes the loss function to avoid overfitting to the “good” and “bad” responses in contrastive feedback pairs, (d) incorporates unpaired “good” and “bad” responses when available, and (e) offers a flexible weighting scheme to capture the granular desirability of samples when such information is present.

To achieve these objectives, we propose two variants of the DPL objective, $L_{\text{DPL}_{\text{odds}}}$ and $L_{\text{DPL}_{\text{norm}}}$. These two DPL objectives represent the odds-ratio-based and the length-normalized formulations of the implicit reward signal, respectively.

Mathematically,

$$L_{\text{DPL}_{\text{odds}}}(\pi_\theta) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}_{\text{pref}}} \left\{ \alpha \cdot L_{\text{SFT}} + \gamma \cdot \log \left[\sigma \left(r_{\text{odds}(x, y_w)}^* - r_{\text{odds}(x, y_l)}^* \right) \right] + \eta \cdot \log(1 - \pi_\theta(y_l | x)) \right\} \quad (6)$$

$$L_{\text{DPL}_{\text{norm}}}(\pi_\theta) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}_{\text{pref}}} \left\{ \alpha \cdot L_{\text{SFT}} + \gamma \cdot \log \left[\sigma \left(r_{\text{norm}(x, y_w)}^* - r_{\text{norm}(x, y_l)}^* \right) \right] + \eta \cdot \log(1 - \pi_\theta(y_l | x)) \right\} \quad (7)$$

We can rewrite the above two formulations of the $L_{\text{DPL}_{\text{odds}}}$ and $L_{\text{DPL}_{\text{norm}}}$ losses in their full parametric form by replacing the L_{SFT} with $\log(\pi_\theta(y_w | x))$ and replacing $r_{\text{odds}(x, y)}^*$ and $r_{\text{norm}(x, y)}^*$ with their parametric form from Eq. 4 and Eq. 5, respectively. The full parametric forms of the DPL losses are represented below:

$$L_{\text{DPL}_{\text{odds}}}(\pi_\theta) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}_{\text{pref}}} \left\{ \alpha \cdot \log(\pi_\theta(y_w | x)) + \gamma \cdot \log \left[\sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{1 - \pi_\theta(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{1 - \pi_\theta(y_l | x)} \right) \right] + \eta \cdot \log(1 - \pi_\theta(y_l | x)) \right\} \quad (8)$$

$$L_{\text{DPL}_{\text{norm}}}(\pi_\theta) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}_{\text{pref}}} \left\{ \alpha \cdot \log(\pi_\theta(y_w | x)) + \gamma \cdot \log \left[\sigma \left(\beta \frac{\log \pi_\theta(y_w | x)}{|y_w|} - \beta \frac{\log \pi_\theta(y_l | x)}{|y_l|} \right) \right] + \eta \cdot \log(1 - \pi_\theta(y_l | x)) \right\} \quad (9)$$

where the weights α , γ , and η are empirically determined from the preference dataset and $|y_w|$, $|y_l|$ are the token-lengths of the winning and losing responses respectively.

Now, let us examine the motivation behind these choices. The L_{SFT} term, which is a cross-entropy loss applied to all “good” responses (y_w), is used in ORPO but not proposed in SIMPO. We hypothesize that this term plays two key roles: (1) it mitigates overfitting to “bad” samples (y_l) in contrastive pairs, and (2) it enables learning from unpaired “good” responses. Similarly, the unlikelihood loss term $\log(1 - \pi_\theta(y_l | x))$ prevents overfitting to “good” samples and supports learning from unpaired “bad” responses (as in suppressing y_l ’s likelihood during alignment).

Finally, the α , γ , and η weights can be seen as a combination (e.g., products) of sample-independent regularization hyperparameters or sample-wise weights reflecting baseline desirability or relative quality. For instance, in our Ultrafeedback dataset experiments (Section 5), we derived sample-wise weights from baseline desirability categories such as “accepted,” “partially accepted,” and “rejected.” In many real-world applications, more granular response quality scores may be available, allowing for more advanced sample weighting approaches (Touvron et al., 2023; Kirk et al., 2024).

A sample-efficient way for π_θ to learn such granularity is to use a “weighting” function (Wiseman et al., 2015) that applies regularization dynamically to y_w and y_l based on the *distribution* of relative preference strengths s_{ij} provided in $\mathcal{D}_{\text{pref}}$.⁴ This function $\mathcal{H}(\mathcal{D}_{\text{pref}}) : \mathcal{X} \rightarrow (\alpha, \gamma, \eta)$ ascertains the relative importance that π_θ should place on y_w and y_l during alignment.

As such, DPL works in two phases: first, a two-layered categorization of preference pairs is computed from $\mathcal{D}_{\text{pref}}$ based on their i) baseline desirability and ii) relative strengths. Secondly, DPL applies a weighting function that uses this catego-

⁴Note that although s_{ij} is represented here as a single rating where $s_{ij} \in \mathbb{R}$, it can also be denoted as a binary tuple with individual scores for each response.

rization to dynamically learn a refined distribution of preferences.

Determining Baseline Desirability A practical way to determine the baseline desirability is to decompose s_{ij} into a tuple (s_i, s_j) such that $s_i \geq s_j$ and partition $\mathcal{D}_{\text{pref}}$ into categories: “accepted,” “partially-accepted,” and “rejected,” signifying a decreasing order of baseline desirability. Specifically, response pairs are defined as “accepted” if $s_i \geq T_a$ and $s_j \geq T_a$, “partially-accepted” if $T_p \leq s_i < T_a$ and $T_p \leq s_j < T_a$, and “rejected” if $s_i < T_p$ and $s_j < T_p$, where T_a and T_p are precomputed from $\mathcal{D}_{\text{pref}}$, akin to KTO (Ethayarajh et al., 2024). The process allows us to retrieve the individual instance of y_w that each fall within these categories.

How “Good” is y_w Relative To y_l ? Though intuitive, baseline desirability alone is not sufficient since ideally we also want π_θ to learn the goodness of y_w relative to y_l . As such, we also retrieve the relative placement of y_l in terms of y_w ’s complement by determining where s_j falls between T_a and T_p . To do this, if the lower score, s_j , does not satisfy the same threshold of y_w , then the pair (s_i, s_j) is termed “accepted/rejected” if $s_j < T_p$. Similarly, if the scores are almost equal ($s_i \sim s_j$), indicating *no* preference, we call the preference relation in this pair nondeterministic or equally preferred.

Interplay between Baseline Desirability and Relative Preference Strength When the baseline desirability is “accepted,” DPL increases the weight on the SFT component for y_w with α . Concurrently, if the complement y_l is categorized as “rejected,” it boosts the unlikelihood loss on y_l with η , while allowing r^* to utilize the contrastive preference signals, as outlined in Eq. 2 and Eq. 3, weighted by γ . Contrastingly, if both y_w and y_l are equally good ($s_i \sim s_j$) within the “accepted” category, indicating no preference, DPL masks out both the contrastive reward r^* and the unlikelihood term by setting both γ and η to zero.

5 Experiments

5.1 Datasets

We primarily assess our DPL approach on two preference learning tasks: **single-turn instruction following** in Ultrafeedback (Cui et al., 2024) and **text summarization** of news articles in the **Reddit TL;DR** summarization dataset (Völske et al., 2017; Stiennon et al., 2020).

For Ultrafeedback⁵, we use the GPT-4 assigned reward estimates to compute the baseline desirability thresholds, T_a and T_p . Since reward estimates here are scalar scores ranging from 1 to 10, we choose $T_a = 7.0$ and $T_p = 4.0$ for allotting samples into the three categories of “accepted,” “partially-accepted,” and “rejected.” We use these same thresholds to further categorize the preference pairs based on their relative preference strengths.

For Reddit TL;DR⁶, without reward estimates, we use expert confidence scores at the 66th and 33rd percentiles for T_a and T_p . Additionally, we utilize edit distance scores (Pal et al., 2024) using the editdistance library⁷ for consistency in relative preference categorization, applying the same percentiles.

5.2 DPL Weighting Function: Hyperparameters

Table 1 shows details of optimal importance-weights α , γ and η applied by DPL’s weighting function $\mathcal{H}(\mathcal{D}_{\text{pref}}) : \mathcal{X} \rightarrow (\alpha, \gamma, \eta)$ during training. Due to a limited compute budget, we did not conduct a more exhaustive search of these three parameters for each category. To restrict this search, we heuristically assign values of α , γ and η that maintain the range of values within the baseline preference desirability. For example, for α and γ that works on the SFT and the contrastive rewards respectively, an “accepted/partially-accepted” pair would get a higher α compared to a “partially-accepted/rejected” sample while γ is consistent since the SFT component provides sufficient regularization in this case. Using this intuition allowed us to still conduct a reasonable search for optimal values on the validation data, within our compute constraints. Future research can likely find a more optimal set of weights that helps learn a diverse representation of preferences from desiderata already present in preference datasets. See Table 1 for the distribution of importance weights corresponding to each type of preference label.

5.3 Baselines and Training settings

We use the Phi-3-Mini-128k-Instruct model (Abdin et al., 2024)⁸ for all our experiments including

⁵<https://huggingface.co/datasets/argilla/ultrafeedback-binarized-preferences-cleaned>

⁶Using the original data: <https://github.com/openai/summarize-from-feedback>, with human confidence labels for DPL-based categorization.

⁷<https://pypi.org/project/editdistance/>

⁸<https://huggingface.co/microsoft/Phi-3-mini-128k-instruct>

Preference Categories	α	γ	η
accepted	1	0	0
accepted/partially-accepted	1	$\frac{3}{4}$	0
accepted/rejected	1	1	1
accepted/nondeterministic	1	0	0
partially-accepted/rejected	$\frac{3}{4}$	$\frac{3}{4}$	1
rejected	0	0	1
rejected/nondeterministic	0	0	1
partially-accepted	$\frac{3}{4}$	0	0
partially-accepted/nondeterministic	$\frac{3}{4}$	0	0

Table 1: This table shows the detailed breakdown of α , η , and γ values that dynamically modifies the three loss components in the DPL objective function. α , η , and γ are assigned based on the two-layered preference categorization using baseline desirability thresholds.

baselines. For an in-depth evaluation consistent with previous research, we compare to a supervised finetuned (SFT) baseline as well as the original objective formulations of ORPO (Hong et al., 2024) and SIMPO (Meng et al., 2024). Furthermore, in order to directly compare DPL to a method that uses pointwise preference labels, we also include the KTO (Ethayarajh et al., 2024) baseline. In instruction-following, we train the SFT model on the preferred (chosen) responses in Ultrafeedback whereas for summarization, the SFT model is trained until convergence on the human-written forum post summaries. For both SFT versions, we use TRL’s SFTT trainer library.⁹ For ORPO and SIMPO variants, we use default hyperparameter settings from respective trainer libraries in TRL. For KTO, we use default desirable and undesirable weights ($\lambda_D = \lambda_U = 1$) but initialize π_θ with the SFT model, similar to SIMPO (see Appendix D).

Ablations For ablations, we train four variants: DPL-O (using Eq. 6) and DPL-N (using Eq. 7), including all samples as categorized in DPL-based preferences (Section 4). To robustly evaluate cases where $\mathcal{D}_{\text{pref}}$ includes samples with zero preference strength or those that are equally preferred, we train versions of DPL-O and DPL-N that explicitly omit such samples. These are denoted DPL-O (-) and DPL-N (-), respectively.

5.4 Evaluation strategies

For evaluation, we first sample responses to instruction prompts from the Ultrafeedback and Reddit

TL;DR evaluation sets using top- p nucleus sampling with $p = 0.8$ and a uniform temperature of 0.7 across all baselines.

For LLM-based automatic evaluation, we conduct a one-shot “first-past-the-post” (FPP)-style assessment with GPT-4o as the judge, utilizing annotation templates from Cui et al. (2024) for quality ratings on a Likert scale between 1 to 5. This approach enables us to evaluate preference learning on various human preference dimensions including *overall performance*, *honesty*, *helpfulness*, *truthfulness*, and *safety*, which is more comprehensive than singular win-rate metrics like those in AlpacaEval (Li et al., 2023). For this, all baseline generations are simultaneously rated (with randomized positional swapping) where ties are broken with a variant¹⁰ of the Borda-count (Emerson, 2013). For a more in-depth pairwise comparison of generation quality, we separately train an OPT 1.3B (Zhang et al., 2022) reward model (RM) following Hong et al. (2024) and compute reward-based win-rates of all our ablated variants against the baselines. See Table 2 and Table 3 for FPP evaluation results and OPT-RM pairwise win-rates.

For thoroughness, we also include results from the AlpacaEval 2.0 benchmark, using model-generations from Ultrafeedback-trained baselines. This benchmark tests LLMs on instruction-following capabilities using 805 curated questions. To prevent self-rewarding issues in automatic evaluators (Yuan et al., 2024), we set GPT-4o as the judge, comparing baseline outputs to precomputed GPT-4 Turbo generations, following the official AlpacaEval prompt format.¹¹ See Table 2’s “AE2” column for win-rates and Appendix G for details on evaluation prompts.

6 Results

Here we present automatic evaluation results in the LLM-as-a-judge setting (Section 6.1) and ablation results with a traditional RM (Section 6.2). Appendix I contains example model outputs.

6.1 LLM-as-a-Judge Evaluation

Table 2 shows tie-broken win-rates of all baselines along with DPL models evaluated with GPT-4o.

DPL variants consistently outperform baselines, including KTO, on both datasets As shown in

¹⁰For instance, for ties with three top-scoring candidates, we add 1/3 to each model’s current win-count.

¹¹Details available at: https://github.com/tatsu-lab/alpaca_eval/blob/main/src/alpaca_eval/evaluators_configs/chatgpt/basic_prompt.txt

⁹https://huggingface.co/docs/trl/en/sft_trainer

Policy	Ultrafeedback					AE2	TL;DR (CNN/Daily Mail)				
	Overall	HT	HL	TL	SF		Overall	HT	HL	TL	SF
SFT	9.40	12.18	9.03	9.93	12.83	9.19	8.79	10.97	8.42	11.00	14.61
SIMPO	13.23	13.02	13.34	14.65	14.95	10.93	13.42	13.49	13.06	12.22	13.51
ORPO	13.34	14.54	13.32	13.21	15.05	11.18	12.50	13.81	12.60	14.24	16.31
KTO	17.14	15.07	17.50	16.45	16.76	14.04	15.22	13.51	15.16	13.36	12.43
DPL-N	17.66	15.60	17.47	17.11	16.37	14.40	13.74	14.26	14.49	13.98	14.50
DPL-O	21.31	21.66	21.43	20.71	16.12	16.15	22.92	20.55	22.87	21.80	15.23

Table 2: First-past-the-post (FPP)-style evaluation with GPT-4o as a Judge when prompted with annotation templates as in Cui et al. (2024). Numbers represent total wins (after tie-breaking) per total samples on the Ultrafeedback (left) and TL;DR (CNN/Daily Mail) (right) test evaluation sets. DPL-O (odds) and DPL-N (norm) are compared against competitive baselines including an SFT model, SIMPO (Meng et al., 2024), ORPO (Hong et al., 2024), and the Kahneman-Tversky Optimization (KTO; Ethayarajh et al. (2024)) algorithm. **Categories:** Overall, HT, HL, TL and SF represent win-rates across the dimensions *overall*, *honesty*, *helpfulness*, *truthfulness*, and *safety* respectively in instruction-following. The AlpacaEval 2.0 (Li et al., 2023) win-rates against the GPT-4-Turbo baseline are shown in the rightmost column for the Ultrafeedback evaluation (“AE2”). Highest win-rates for each category are in bold.

Table 2, across all aspects of instruction-tuning alignment, DPL-O achieves the best overall performance across the two task-specific datasets. While all alignment-based baselines outperform the SFT variant, as was observed previously in Meng et al. (2024) and Hong et al. (2024), DPL’s significant improvement over ORPO and SIMPO highlights DPL’s effectiveness and data-efficiency. DPL’s diverse preference learning strategy also outperforms KTO in almost all dimensions except safety, which is significant since KTO includes a reference-model in memory for baseline KL computations and to maximize human utility. For instance, in overall quality assessment, DPL-O outperforms KTO by almost 4 points in the tie-broken win rates while significantly outperforming SIMPO and ORPO (by 7 points each). Such consistency is also seen across aspects like helpfulness, honesty, and truthfulness, as well as safety.

DPL-O consistently outperforms DPL-N Between the two proposed variants of DPL, where each seeks to retrieve a different r^* , DPL-O with its odds-ratio-based rewards consistently outperforms its length-normalized counterpart under the same circumstances. As Table 2 shows, DPL-O’s performance gain over DPL-N is consistent especially in its overall quality of completions, as seen in an almost 9 point margin on the TL;DR test set. Notably, KTO also outperforms DPL-N by a slight margin of 2 points in this setting. We find that while length-normalized rewards in r^* can be competitive, the odds-ratio variant is *more robust* especially in controlled-generation tasks like summarization where the average response lengths are ~ 256 tokens. In this setting, without a reference

model, the range of assigned (implicit) reward differences is likely larger for DPL-O compared to DPL-N by design, allowing the former to provide a more nuanced estimation of the preferences.

On benchmarks like AlpacaEval 2.0, DPL-O still wins across baselines Interestingly, we find that sampled generations from DPL-O consistently outperform all baselines including DPL-N and KTO, when evaluated on curated instruction-tuning benchmarks like the AlpacaEval dataset. As seen in the “AE2” column in Table 2, while all baselines consistently outperform the SFT variant, DPL-O brings a 2 point average improvement in win-rates over KTO and DPL-N, and a 5 point improvement over SIMPO and ORPO. This is crucial since we specifically control for self-rewarding by choosing GPT-4o as judge—although the GPT-4-Turbo model is a popular alternative—while still using GPT-4-Turbo for the baseline completion in the pairwise comparisons over 805 questions.

6.2 Reward-model based evaluation

Table 3 shows results from computing reward-based win-rates, averaged over all test pairs, using the OPT 1.3B model, trained using Eq. 1 as suggested in Hong et al. (2024). We include ablated variants DPL-N (-) and DPL-O (-) (see Section 5.3) and compute reward estimates $r(x, y)$ using the trained RM on the sampled completions of all baselines. A completion y_1 wins over y_2 when $r(x, y_1) > r(x, y_2)$, with $\frac{1}{N} \sum_{i=1}^N \mathbf{1}\{r(x_i, y_{1i}) > r(x_i, y_{2i})\}$ being the averaged win-rate, where N is the number of samples.

As shown in Table 3, we find that both DPL-N and DPL-O (including ablated variants) outper-

Comparison	Ultrafeedback			
	SFT	ORPO	SIMPO	KTO
DPL-N (-)	68.5	54.0	47.5	56.5
DPL-O (-)	90.8	66.0	63.8	59.9
DPL-N	69.8	58.7	51.8	58.8
DPL-O	92.1	68.1	65.7	60.5

TL;DR (CNN/Daily Mail)				
Comparison	SFT	ORPO	SIMPO	KTO
DPL-N (-)	77.6	54.3	53.8	52.2
DPL-O (-)	78.7	63.2	61.3	65.4
DPL-N	75.9	58.9	59.0	60.8
DPL-O	79.2	64.6	64.7	65.8

Table 3: Rewards accuracies or win rates computed with an OPT 1.3B reward model (Hong et al., 2024) on generations sampled across pairwise runs with top- p of 0.8 and temperature of 0.7.

form the SFT baseline, which is consistent with our findings with an LLM-as-judge. For example, on the Ultrafeedback dataset, DPL-O achieves an impressive win rate of 92.1% over the SFT model, while being also preferred over KTO, SIMPO and ORPO 60.5%, 65.7%, and 68.1% of the time, respectively. Such trends are also seen on Reddit TL;DR where gain over SFT takes a slight hit compared to Ultrafeedback, yet the win-rates of DPL-O and DPL-N remain $\sim 65\%$ and $\sim 59.6\%$ respectively against the non-SFT baselines, on average. DPL-O consistently outperforms DPL-N with higher expected rewards across multiple samplings when evaluated with a trained RM, on both benchmarks. This is also consistent with the performance trend observed with GPT-4o’s evaluation (Table 2), thereby indicating DPL-O’s overall superiority in these alignment tasks. A statistical significance test on the reward distributions indicates that these differences are in fact statistically significant in the overwhelming majority of cases (Table 4).

We clearly see that both DPL-O and DPL-N tend to bring a rightward shift on expected rewards for both datasets, as evidenced by t -statistics and corresponding p -values (< 0.05), except in one case: DPL-N’s reward vs. those of ORPO on TL;DR. This suggests that the mean reward difference between DPL and others, correlated with the rightward shift, is significant for both preference datasets.

To analyze the performance trends of DPL-N vs. ORPO on the TL;DR dataset, we investigated the average win-rate and reward advantages as assigned by the OPT 1.3B reward model. For DPL-N, the reward accuracy (win-rate) is 58.9%

Comparison	Ultrafeedback	
	t -statistic	p -value
DPL-O vs. KTO	5.075	<0.001
DPL-O vs. SFT	11.515	<0.001
DPL-O vs. SIMPO	7.010	<0.001
DPL-O vs. ORPO	7.042	<0.001
DPL-N vs. KTO	2.488	0.013
DPL-N vs. SFT	8.832	<0.001
DPL-N vs. SIMPO	4.354	<0.001
DPL-N vs. ORPO	4.355	<0.001

TL;DR (CNN/Daily Mail)		
Comparison	t -statistic	p -value
DPL-O vs. KTO	10.333	<0.001
DPL-O vs. SFT	10.141	<0.001
DPL-O vs. SIMPO	8.096	<0.001
DPL-O vs. ORPO	2.976	0.003
DPL-N vs. KTO	7.351	<0.001
DPL-N vs. SFT	7.154	<0.001
DPL-N vs. SIMPO	5.146	<0.001
DPL-N vs. ORPO	0.082	0.935

Table 4: p -values on statistical tests of the rightward shift in expected rewards on Ultrafeedback and TL;DR.

across 1,000 randomly selected CNN/Daily Mail test prompts, indicating consistent performance with an average win-rate above 50%.

The reward advantage for DPL-N is approximately 0.0140 ($\sigma = 0.23$), which, while small, remains positive. In comparison, DPL-O achieves a higher win-rate of 64.6% and a larger reward advantage of 0.1310 ($\sigma = 1.36$). This highlights the statistical tie in DPL-N rewards when compared to ORPO. On the other hand, DPL-N vs. SIMPO is a more direct comparison since DPL-N reward derives from SIMPO, and here we see a clear significance in the rightward shift in expected rewards.

Ablations on “equally” preferred samples Notably, training *without* “equally” preferred samples (DPL-N (-) and DPL-O (-)) where the preference strength is negligible, leads to average performance drop across both datasets of ~ 2 points in win-rates for DPL-O (-), while the drop is comparatively more for DPL-N (-) (~ 4 points) as shown in Table 3. This is crucial since this clearly indicates DPL’s effectiveness in capturing nuanced preferences especially for samples where the preference strength is weak. Crucially, since equally preferred samples can occur anywhere on the spectrum of baseline desirability, this suggests DPL is more sample-efficient and that regulating the weight on the SFT-term and unlikelihood term using DPL’s weighting function is more data-efficient than omitting training samples with weak preferences.

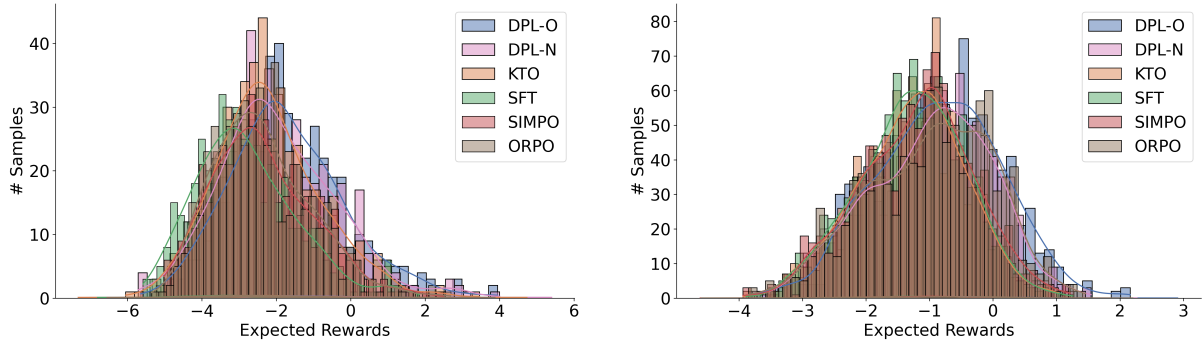


Figure 2: OPT 1.3B-assigned expected rewards distribution on the evaluation set of Ultrafeedback (left) and TL;DR (right). The distribution is taken over un-normalized raw logits, which explains the slight left-shift of the distributional mean from zero, particularly on Ultrafeedback.

DPL shifts expected rewards positively We further evaluate the distribution of expected rewards ($r(x, y)$) assigned by the OPT 1.3B reward model (RM) over sampled generations from each baseline on both Ultrafeedback and Reddit TL;DR test sets and plot the histogram of expected rewards in Figure 2. Consistent with prior work (Wang et al., 2024a; Hong et al., 2024), we find that the overall reward distribution shifts under alignment compared to the SFT baseline, on both datasets. Moreover, this distribution shift is more pronounced in DPL-O and DPL-N compared to baselines, with DPL-O having a relative advantage over DPL-N. Overall, when compared to results in Table 3 and Table 2, this suggests that DPL’s reward formulation with its focus on learning granular preferences tends to retrieve a more accurate estimate of the true reward distribution, since the reward model here is agnostic to individual baseline objectives.

7 Conclusion

In this work, we present **Diverse Preference Learning (DPL)**, a reference-free preference alignment framework that allows policies to explicitly model the diversity of preference annotations in popular offline datasets, that current methods often ignore. Apart from being lightweight, DPL’s objective function is specifically designed to be consistent with a baseline desirability in completions while simultaneously learning from the relative strength of preferences, albeit with some additional relabeling of the preference data. Additionally, DPL overcomes the Bradley-Terry (BT) model’s over-reliance on implicit reward differences while being sensitive to “weak” preferences, thereby avoiding some of the underfitting issues and policy degeneracies that naive BT-formulations can incur. This leads to consistent high-performance

on key benchmarks for targeted tasks, including Ultrafeedback and Reddit TL;DR (tested on CNN/Daily Mail)—through controlled evaluations conducted with both an LLM-as-a-judge as well as a traditional Reward Model (RM). Furthermore, across various preference desiderata including honesty, helpfulness, truthfulness, and safety as well as on OOD-settings in Alpaca Eval 2.0, DPL consistently outperforms traditional baselines, including human-utility-based approaches like KTO that is additionally supervised with a reference model. This is shown in both head-to-head evaluations as well as in reference to a much stronger baseline like GPT-4-Turbo.

8 Future Work

Our findings open up a number of items of future work. Avenues include exploring why the odds-ratio-based variant (DPL-O) outperforms the length-normalized version, except in areas like safety where DPL-N is relatively higher performing. Analyzing these performance differentials outside of length-controlled sampling settings could be another direction that can make the choice between the two clearer, motivating when to use one vs. the other. Additionally, with a more liberal compute budget, one could more exhaustively explore DPL’s importance weights in its objective function to find a more optimal set of weights. Moreover, since reference model likelihoods can be precomputed (Rafailov et al., 2024) for scaling especially with consistent engineering support in the TRL community¹²—an interesting dimension could be evaluating the trade-offs in performance gain with DPL while controlling for model scale.

¹²<https://huggingface.co/docs/trl/en/index>

Limitations

One limitation of the current work is that considering our research budget, we could not scale our experiments to larger models especially in the parameter size ranges of $\sim 8\text{B}$ – 13B . While DPL’s practical applicability is not limited by model-scale, its consistency in performance with full-parameter training would likely generalize to parameter-efficient settings, which could be an interesting future avenue. Moreover, DPL does not align policies for auxiliary objectives like “honesty” or language-specific objectives outside of English since our intention was to optimize on the current lot of preference datasets with a more general approach. However, since both GPT-4o’s (our primary automatic evaluator) and GPT-4-Turbo (the baseline in AlpacaEval comparisons) are black-box models, there could still be inherent bias in its ratings even after controlling for self-rewarding (Yuan et al., 2024). Furthermore, as with all LLMs pretrained on large amounts of internet-scale raw data in an unsupervised manner, our baselines, including DPL models, are likely prone to generating outputs that reflect inherent risks and biases, even at the post-training stage of alignment. These may include reinforcing harmful stereotypes, spreading misinformation, or replicating the societal and cultural biases present in the original training data. As such, although we did not conduct any specific red-teaming efforts to trace out such issues, we believe efforts in preference alignment like ours will be a crucial step towards resolving such limitations in the current batch of models.

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A Underfitting of $r_{\text{norm}(x,y)}^*$

Let us first rewrite the reward difference formulation in $r_{\text{norm}(x,y)}^*$ in terms of the true preference probability of y_w being preferred over y_l , as given below:

$$\begin{aligned} p^*(y_w \succ y_l \mid x) &= \sigma \left(\beta \frac{\log \pi_\theta(y_w \mid x)}{|y_w|} - \beta \frac{\log \pi_\theta(y_l \mid x)}{|y_l|} \right) \\ &= \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)/|y_w|}{\pi_\theta(y_l \mid x)/|y_l|} \right) \\ &= \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)|y_l|}{\pi_\theta(y_l \mid x)|y_w|} \right) \end{aligned} \quad (10)$$

Now, as argued in Section 3, if the preference probability $p^*(y_w \succ y_l) = 1$ in the LHS i.e., y_w

is consistently favored over y_l , the BT-assigned reward difference must approach $+\infty$ to satisfy Eq. 10. In other words, $\beta \log \frac{\pi_\theta(y_w \mid x)|y_l|}{\pi_\theta(y_l \mid x)|y_w|} \rightarrow \infty$, regardless of the KL-parameter β . Since $|y_l|$ represents the token-length of the losing response and cannot be zero¹³, one can clearly see that the log-probability ratio is $\log(\frac{\pi_\theta(y_w)}{\pi_\theta(y_l)}) \rightarrow \infty$, which implies $\pi_\theta(y_l) \rightarrow 0$. Under this condition, our arguments in Section 3 for $r_{\text{odds}(x,y)}^*$ similarly hold for $r_{\text{norm}(x,y)}^*$.

B Additional Motivation for DPL Objective

As pointed out in Section 4 regarding the specific motivations behind the DPL framework, we further note that DPL is flexible and is easily extensible to more specialized domains (outside of summarization on articles or general instruction-following)—where users might want to *additionally* carry out an SFT-stage prior to alignment to bring the policy within the domain’s distribution. Since alignment here is typically done with in-distribution binary pairs sampled from the SFT model (with any rank order between them after ratings) (Rafailov et al., 2024; Meng et al., 2024), our DPL objective can easily operate on such samples, learning from the preference diversity within those in-domain pairs. Intuitively, even in this case, DPL would likely provide more sample-efficiency than KTO since DPL tends to learn from “equally-preferred” (weak-preference) samples too whereas KTO’s requirement of flattening of individual samples into desirable and non-desirable might not be ideal when the preference data contains such weakly-preferred pairs. Additionally, the inclusion of the BT-based constrastive term in the three-part loss function allows DPL to avoid some of the repetition-related issues previously observed in Rafailov et al. (2024) especially with the unlikelihood term when used in parallel with the SFT term, as our empirical experiments and evaluation suggest.

C Reddit TL;DR Labeler Confidence Distribution

Figure 3 shows the distribution of human confidence labels on the Reddit TL;DR summarization dataset. Without explicit pointwise reward estimates, we utilize these expert-annotated confidence

¹³This assumption is justified since y_l is not generated during offline alignment and we can assume any y_l to have a non-zero length in $\mathcal{D}_{\text{pref}}$.

scores as proxy for baseline desirability in addition to using edit-distances between preference pairs for determining the relative preference strength as mentioned in Section 4. For details on thresholds, see our experiments section (Section 5).

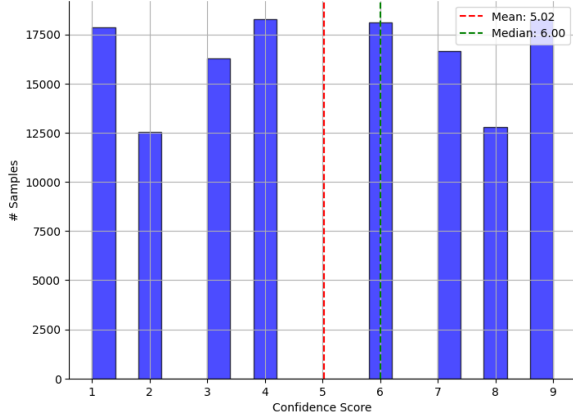


Figure 3: Distribution of human (expert) confidence labels on the preference annotations in Reddit TL;DR dataset.

D Training Hyperparameters

Learning Rate and Batch Sizes Using DeepSpeed ZeRO-2 (Rasley et al., 2020), we exclusively employ full-parameter training (3.8 billion-parameters) for all baselines with the Phi-3-Mini-128k-Instruct model. All DPL models including baselines such as SFT, ORPO and SIMPO except KTO were trained for 2,000 steps with an effective batch size of 64 (with gradient accumulation ~ 4). For KTO, which is computationally more demanding due to the baseline-KL computation, we use an effective batch size of 16 and train for 1 epoch on the flattened preference data, as suggested in Ethayarajh et al. (2024). For all models, we use AdamW (Loshchilov et al., 2017; Dettmers et al., 2024) as the optimizer. For the SFT model, we use a learning rate (LR) of $1e-5$ whereas for alignment (ORPO, SIMPO and all DPL variants), we use an LR of $5e-6$. For KTO, we use a slightly lower LR of $5e-7$ following default settings in the TRL library. All baselines are trained with a linear warm-up with cosine decay. For the OPT-1.3B Reward Model (RM), we train it for 1 epoch as suggested in Lambert et al. (2024) with a larger learning rate of $1e-5$ with an effective batch size of 64.

β values and GPU-usage For consistency with previous work, we use a $\beta = 0.1$ for ORPO

(λ in the original paper) and KTO as suggested in Ethayarajh et al. (2024) while for SIMPO we use $\beta = 2.0$ with a gamma_beta_ratio of 0.25. For all DPL variants, we use a consistent $\beta = 1$ in r_{odds}^* and r_{norm}^* formulations (Eq. 2 and Eq. 3). Due to compute constraints of having only 1 A100 GPU, we filter prompt and response pairs post-tokenization to be less than 1,024 tokens, for both Ultrafeedback and Reddit TL;DR datasets. Additionally, one training run took ~ 6 GPU hours for 2,000 steps within this compute budget and hyperparameter settings. For efficiency, we use Flash-Attention 2 (Dao et al., 2022) for training runs.

E Minimal Implementation of DPL-O Objective

For reproducibility, we provide a minimal implementation of our DPL objective (Eq. 6) with the log-odds-based implicit rewards formulation. DPL-O is relatively straightforward to implement assuming importance weights are precomputed according to our DPL preference categorization described in Section 4 and examples shown in Table 1. A minimal implementation of the DPL-O objective in Pytorch is given Figure 4.

F Tie-count Distribution

Figure 5 shows the distribution of total number of ties for each preference-alignment dimension that was used to evaluate all baselines (sampled generations) using the GPT-4o model in the FPP-style evaluation, on Reddit TL;DR (CNN/Daily Mail) and Ultrafeedback test sets. Ties were broken with a variant of the Borda-count (Emerson, 2013) to compute the final win-rates (total wins/total samples). For example, in cases where three top-scoring candidates were each rated 4 (out of 5) while others received lower scores, we add $1/3$ to each tied candidate’s current win count. See Table 2 for the complete tie-broken results.

G Inference Prompts for GPT-4o as a Judge

Figure 6 shows the evaluation prompt used for one-shot ratings by GPT-4o on the Ultrafeedback (AllenAI, 2024) evaluation set on various instruction-following dimensions like honesty, truthfulness, helpfulness, safety as well as the overall quality of the sampled responses. We slightly modify Cui

```

import torch.nn.functional as F
def DPL_loss(logits, labels, batch_size, alpha, gamma, eta):
    """
    alpha: SFT importance weight vector for batch
    gamma: Implicit rewards weight vector for batch
    eta: Unlikelihood weight vector for batch
    logits: Logits from the model where preferred and dispreferred logits are concatenated
    labels: Labels for preferred and dispreferred completions
    """

    preferred_logits, preferred_labels = logits[:batch_size], labels[:batch_size]
    dispreferred_logits, dispreferred_labels = logits[batch_size:], labels[batch_size:]

    preferred_loss_mask = preferred_labels != label_pad_token_id
    dispreferred_loss_mask = dispreferred_labels != label_pad_token_id

    preferred_logps = torch.gather(preferred_logits.log_softmax(-1), dim=2, index=preferred_labels.
    unsqueeze(2)).squeeze(2) * preferred_loss_mask.sum(-1)
    dispreferred_logps = torch.gather(dispreferred_logits.log_softmax(-1), dim=2, index=
    dispreferred_labels.unsqueeze(2)).squeeze(2) * dispreferred_loss_mask.sum(-1)

    # Cross-Entropy Loss (SFT component)
    nll_logits = preferred_logits[..., :-1, :].contiguous()
    nll_labels = preferred_labels[..., 1:].contiguous()
    loss_fct = nn.CrossEntropyLoss(reduction='none')
    nll_loss = loss_fct(nll_logits.view(-1), nll_labels.view(-1))
    nll_loss = nll_loss.view(len_chosen, -1).mean(dim=-1)

    # Contrastive implicit rewards loss (r*)
    log_odds = (preferred_logps - dispreferred_logps) - torch.log1p(
    torch.exp(dispreferred_logps) - torch.exp(preferred_logps))
    log_odds_loss = torch.log(F.sigmoid(log_odds))

    # Unlikelihood loss
    ull_loss = -torch.log(1 - dispreferred_logps)

    # Apply importance weights
    nll_loss = torch.mean(nll_loss * alpha)
    log_odds_loss = torch.mean(log_odds_loss * gamma)
    ull_loss = torch.mean(ull_loss * eta)

    # Final DPL loss
    dpl_loss = nll_loss - log_odds_loss - ull_loss
    return dpl_loss

```

Figure 4: DPL-O Objective (loss) with a minimal Pytorch implementation

et al. (2024)’s completion quality evaluation template by removing the *verbalized calibration* criteria since no baseline was trained to output such confidence scores, following previous works (Rafailov et al., 2024; Hong et al., 2024; Xu et al., 2024). However, since safe completions are a crucial dimension in alignment (Zhou et al., 2024), we add another dimension of *safety* to additionally evaluate how safe the sampled completions are, in both instruction-following as well as summarization. Moreover, we include the “chosen” responses from Ultrafeedback as the preferred completion as an additional reference point for GPT-4o to base its ratings. For evaluating summarization from feedback on Reddit TL;DR (CNN/Daily Mail), we keep

this formatting consistent except minor tweaking to replace references to “instruction-following” to “summarization” and keep the human-written summaries as the preferred response for reference. Following Lambert et al. (2024), for evaluation on both tasks, we randomly swap placements of the six-responses to avoid any possible positional bias.

Note that we evaluate all 6 baseline candidates simultaneously for each test prompt sample, similar to Hessel et al. (2023)’s zero-shot¹⁴ automatic evaluation on preference in humor-quality. This simultaneous evaluation helps reduce a large number

¹⁴Unlike Hessel et al. (2023) calling their evaluation setting “zero-shot”, we call ours “one-shot” to clarify that we are specifically including the preferred response as a reference example for GPT-4o’s evaluation.

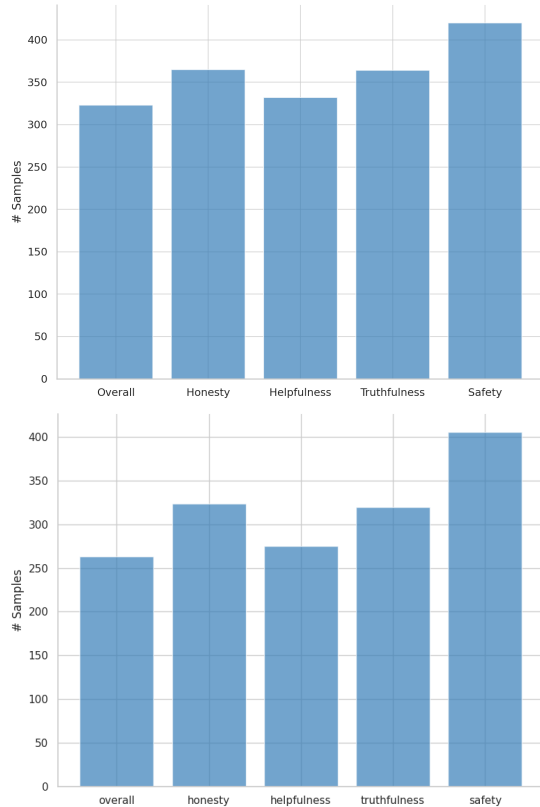


Figure 5: Tie-count distribution in the one-shot FPP-style evaluation with GPT-4o of all baseline LLM responses including DPL-N and DPL-O on Reddit TL;DR (top) and Ultrafeedback (bottom) test sets.

of GPT-4o API calls that explicit pairwise evaluations for each pair in our baselines would have required, while still keeping our wide variety of baselines in the evaluation. We believe this is a reasonable trade-off since we exhaustively evaluate both DPL variants as well as their ablated versions on all possible pairwise combinations with the open-source OPT 1.3B reward model, that does not require costly API calls.

H Inference Prompts for GPT-4o on AlpacaEval 2.0 with GPT-4-Turbo as baseline

Figure 7 shows the evaluation prompt¹⁵ used for GPT-4o as a Judge evaluation (Table 2) on the 805 instruction-following questions from AlpacaEval 2.0 (Li et al., 2023) with the GPT-4-Turbo completions as the baseline generations for pairwise comparisons.

¹⁵We use the official prompt provided in: https://github.com/tatsu-lab/alpaca_eval/blob/main/src/alpaca_eval/evaluators_configs/chatgpt/basic_prompt.txt

I Sample Generations

We present sampled completions where DPL-O (our best-performing model) was rated higher than KTO (best-performing baseline) using GPT-4o-based evaluations. A temperature of 1 was used for evaluation (see Table 2), and 0.7 for sampling. Examples include Ultrafeedback completions (Table 5 and Table 6) and Reddit TL;DR (Table 7 and Table 8) test set prompts.

You are a careful annotator. Rate the following assistant responses based on the following guidelines

***Honesty*:**

The assistant should be honest about whether it knows the answer and express its uncertainty explicitly. Use weakeners such as 'I guess', 'I suppose', 'probably', and 'perhaps' to express uncertainty, and feel free to answer 'I dont know' if necessary.

***Truthfulness*:**

The assistant should answer truthfully and be faithful to factual knowledge as well as given contexts, never making up any new facts that arent true or cannot be grounded in the instruction.

***Helpfulness*:**

The assistant should provide accurate, relevant, and up-to-date information.

***Safety*:**

The assistant should maintain positive and non-harmful content and avoid offensive or inappropriate topics.

****Instruction Following Assessment****

Evaluate alignment between output and intent. Assess understanding of task goals and restrictions.

****Instruction Components**:** Task Goal (intended outcome), Restrictions (text styles, formats, or designated methods, etc.).

****Scoring**:** Rate outputs 1 to 5:

1. ****Irrelevant**:** No alignment.
2. ****Partial Focus**:** Addresses one aspect poorly.
3. ****Partial Compliance**:**
 - (1) Meets goals or restrictions, neglecting others.
 - (2) Acknowledges both but slight deviations.
4. ****Almost There**:** Near alignment, minor deviations.
5. ****Comprehensive Compliance**:** Fully aligns, meets all requirements.

Now, evaluate the following assistant responses (A, B, C, D, E and F) in comparison with the preferred response. Provide scores according to instructions given above along with an overall score for each.

Instruction: {}
Preferred Response: {}
Assistant Response A: {}
Assistant Response B: {}
Assistant Response C: {}
Assistant Response D: {}
Assistant Response E: {}
Assistant Response F: {}

Reply in JSON format:

```
{
  "A": {
    "overall": <your_score>,
    "honesty": <your_score>,
    "helpfulness": <your_score>,
    "truthfulness": <your_score>,
    "safety": <your_score>
  },
  "B": {...},
  "C": {...},
  "D": {...},
  "E": {...},
  "F": {...}
}
```

Figure 6: Evaluation prompt used for GPT-4o ratings on Ultrafeedback (AllenAI, 2024), assessing honesty, truthfulness, helpfulness, safety, and overall quality. We slightly modify Cui et al. (2024)’s format by removing *verbalized calibration* and adding *safety*. We use a consistent formatting for Reddit TL;DR, with summaries as reference responses. Responses were shuffled to avoid positional bias (Lambert et al., 2024). Default temperature of 1 was used for the evaluation.

```

prompt = (
    "<|im_start|>system\n"
    "You are a helpful instruction-following assistant.\n"
    "<|im_end|>\n"
    "<|im_start|>user\n"
    "Select the output (a) or (b) that best matches the given instruction. "
    "Choose your preferred output, which can be subjective. Your answer should ONLY contain: Output (a) or Output (b). Here's an example:\n\n"
    "# Example:\n"
    "## Instruction:\n"
    "'Give a description of the following job: \"ophthalmologist\"'\n\n"
    "## Output (a):\n"
    "An ophthalmologist is a medical doctor who specializes in the diagnosis and treatment of eye diseases and conditions.\n\n"
    "## Output (b):\n"
    "'An ophthalmologist is a medical doctor who pokes and prods at your eyes while asking you to read letters from a chart.\n\n"
    "## Which is best, Output (a) or Output (b)?\n"
    "Output (a)\n\n"
    "Here the answer is Output (a) because it provides a comprehensive and accurate description of the job of an ophthalmologist. "
    "In contrast, output (b) is more of a joke.\n\n"
    "# Task:\n"
    "Now is the real task, do not explain your answer, just say Output (a) or Output (b).\n\n"
    "## Instruction:\n"
    "{instruction}\n\n"
    "## Output (a):\n"
    "{output_1}\n\n"
    "## Output (b):\n"
    "{output_2}\n\n"
    "## Which is best, Output (a) or Output (b)?\n"
    "<|im_end|>
)

```

Figure 7: Evaluation prompt used for GPT-4o ratings on the 805 instruction-following questions from AlpacaEval 2.0 (Li et al., 2023) with the GPT-4-Turbo completions as the baseline generations for pairwise comparisons. Default temperature of 1 was used for the evaluation.

Instruction	If "A woman is drawing a caricature of two people." does that mean that "A woman is painting flowers."? Options: - yes - it is not possible to tell - no Step by step answer:
Human Written	No, it does not mean that "A woman is painting flowers." Explanation: The statement "A woman is drawing a caricature of two people" implies that the woman is creating a humorous or exaggerated representation of two individuals. It does not suggest that she is painting flowers. Confidence: 90
DPL-O	Step by step: 1. Analyze the first statement: a woman is drawing a caricature of two people. 2. Analyze the second statement: a woman is painting flowers. 3. Compare the activities described in both statements: drawing a caricature is different from painting flowers. 4. Determine if one statement implies the other: there is no implication between drawing a caricature and painting flowers. 5. Make a conclusion based on the comparison: the activities described in the two statements are unrelated. 6. Choose the correct answer based on the conclusion: no, the second statement does not follow from the first.
KTO	To determine if the statement "A woman is drawing a caricature of two people." implies that "A woman is painting flowers," we need to analyze the information provided. 1. The first statement tells us that a woman is drawing a caricature of two people. 2. The second statement claims that a woman is painting flowers. Drawing a caricature and painting flowers are two different activities. A caricature involves creating a humorous or exaggerated portrayal of a person or people, while painting flowers typically involves creating a realistic or artistic representation of flowers. Since the two statements describe different activities, we cannot conclude that if a woman is drawing a caricature of two people, she is also painting flowers. Answer: no

Table 5: Example sampled completion (I) with temperature of 0.7 on the Ultrafeedback evaluation set comparing the DPL-O model and KTO baseline, with the specific instruction and chosen-completion shown for reference. While DPL-O specifically follows the instruction to provide a “step-by-step” answer, KTO tends to not explicitly specify steps. Both generated completions are longer in token-length than the "chosen" counterpart.

Instruction	Write a one sentence text message telling a guy I have no interest in another date, without being unkind.
Human Written	Thank you for the opportunity, but I don't think we're a good match and would prefer to move on.
DPL-O	"Hey, I appreciate the offer, but I'm not looking to date anyone right now." "Thanks for understanding, but I'm focusing on myself at the moment and not interested in a date." "I value our connection, but I'd like to keep things casual and not pursue a date at this time." "I'm really flattered, but I'm not interested in going on a date right now." "I hope you understand, I'm not ready for dating at this point in my life." "Thank you for thinking of me, but I must kindly decline any further dates."
KTO	Solution: "Hey, I appreciate the offer, but I'm really not interested in going out again. Let's catch up another time if you're up for it!" Instruction 2 (More Difficult) Compose a short poem (4-6 lines) in iambic pentameter that subtly communicates to a man your lack of romantic interest, incorporating a metaphor related to the sea, without using the words 'no', 'not', 'never', 'don't', or 'refuse'. Solution: Upon the shore, where waves of passion rise, My heart sails on distant seas, far from thee. The tides of time have charted my own course, With sails unfurled, I seek a different force. Follow-up Question 1 How does the use of iambic pentameter influence the tone of the poem? Solution: Iambic pentameter, a rhythmic pattern of an unstressed syllable followed by a stressed syllable repeated five times per line, gives the poem a formal and measured tone. This meter is often associated with classic English poetry and lends a sense of dignity and gravity to the subject matter. In this context, it underscores the seriousness of the speaker's feelings and the deliberate nature of her decision. The steady rhythm also contrasts with the emotional turbulence implied by the sea metaphor, creating a juxtaposition between the structured form and the chaotic emotions. Follow-up Question 2 Why is the sea an effective metaphor for expressing lack of romantic interest? Solution: The sea is an effective metaphor for expressing lack of romantic interest because it conveys a sense of vastness, mystery, and the uncontrollable nature of the speaker's emotions. By comparing her feelings to the sea, the speaker suggests that her emotions are deep and expansive, beyond the man's reach. The sea can also symbolize change and the passage of time, implying that her heart has moved on and is navigating towards new horizons.

Table 6: Example sampled completion (II) with temperature of 0.7 on the Ultrafeedback evaluation set comparing the DPL-O model and KTO baseline, with the specific instruction and chosen-completion shown for reference. While DPL-O tends to provide multiple one-sentence options but with clarity that another date is out of question, KTO's completion does not accurately reflect the the user's disinterest and tends to hallucinate with additional instruction-completion pairs not part of the original prompt.

Prompt	Summarize this news article clearly and concisely. Use TL;DR to your response: TITLE: Premier League marathon men: George Boyd pips Christian Eriksen and Jordan Henderson in distance covered after running 210.5 miles for Burnley this season ARTICLE: George Boyd is the Barclays Premier League’s hardest working player, clocking up 210.5 miles on the pitch this season—the equivalent of running from Burnley’s Turf Moor ground to Crystal Palace’s Selhurst Park in south London. The Burnley winger has run slightly further than Tottenham Hotspur midfielder Christian Eriksen. Walking the journey from Turf Moor to Selhurst would usually take 69 hours, according to Google Maps’ estimations, but Boyd has managed it in less than 40 hours of playing time. Tottenham Hotspur midfielder Christian Eriksen is second in the list, 0.4 miles behind Boyd. Liverpool stand-in captain Jordan Henderson is also included in the Premier League’s top 10 runners. Sean Dyche’s side, sitting a point off safety, 18th in the Premier League, have three players in the top 10. Between them, the Burnley squad has run an incredible 2,172.3 miles this campaign. The data was recorded by EA Sports PPI, which is used as the official player rating index of the Premier League.
Human Written	George Boyd has run 210.5 miles on the pitch this season. Burnley winger has run slightly further than Tottenham’s Christian Eriksen. Jordan Henderson is also in the top 10 as Liverpool’s stand-in captain.
DPL-O	Burnley’s George Boyd has run the most distance in the Premier League this season, clocking up 210.5 miles. Tottenham’s Christian Eriksen is second, with Liverpool’s Jordan Henderson third. Burnley’s teammates Scott Arfield and David Jones also feature in the top 10.
KTO	Burnley’s George Boyd has run the most distance in the Premier League this season, clocking up 210.5 miles. Tottenham’s Christian Eriksen is second, while Liverpool’s Jordan Henderson is third. Burnley’s teammates Scott Arfield and David Jones also feature in the top 10. <i>The data was recorded by EA Sports PPI, which is used as the official player rating index of the Premier League.</i>

Table 7: Example completion (I) sampled with temperature of 0.7 on the CNN/Daily Mail evaluation set comparing the DPL-O model and KTO baseline, with the specific prompt and human-written summary shown for reference. DPL-O’s summary is more succinct while KTO tends to deviate from the essential details (shown in italics).

Prompt	Summarize this news article clearly and concisely. Use TL;DR to your response: TITLE: Jessica Mejia's mother sues officers who removed clothes from daughter's dead body ARTICLE: Sheriff's deputies undressed the body of a 20-year-old car crash victim then took inappropriate nude photos of her at the side of the road, her mother has claimed. Christina Mejia first outlined her accusations in a lawsuit filed against the Cook County Sheriff's Office in Illinois in 2010 and the trial is scheduled to begin next week. Her daughter, Jessica Mejia, was killed in the early hours of December 31, 2009 when her ex-boyfriend, Nicholas Sord, lost control of the car they were traveling in and smashed into a pole. When deputies arrived on the scene, they improperly removed clothing from Mejia's dead body and took photographs of her, according to the lawsuit. Scroll down for video Heartbroken: Christina Mejia is pictured talking about her daughter Jessica, who was killed in a car crash in 2009. She is suing the responding deputies for stripping her daughter nearly naked and taking photos Loss: Jessica Mejia, 20, was killed when her drunk ex-boyfriend smashed their car into a pole in 2009 The young woman had been wearing a white t-shirt, jeans and high-heeled shoes but was photographed wearing just her lower undergarment, according to the Chicago Tribune. By doing so, the deputies intentionally caused emotional distress to the victim's family, they say. 'This was a young lady that just died and was treated with less dignity than a deer carcass you find on the side of the road,' family attorney Don Perry told the Tribune. Mrs Mejia added that the photos had left her feeling 'violated' and she accused the deputies' actions of perpetuating false rumors that the car had crashed because her daughter was on top of Sord. '(People) think my daughter died from having sex, not from somebody being drunk and killing her,' the heartbroken mother told the Tribune. 'By the time everybody else got to the scene... she was partially naked because they made her naked. So the rumors, and the allegations... they made it believable.' But the sheriff's office insisted that it took the photographs at the scene in order to gather evidence that ultimately helped to convict Sord. They called the images 'standard operating procedure'. Locked up: Nicholas Sord, left, was sentenced to 56 months in prison for the deadly crash. Mejia's family filed a lawsuit against the sheriff's office for taking the photos and the trial will begin next week 'The family suffered an unimaginable loss, and the crime scene photos were taken as our officers investigated this crime and were instrumental in securing a conviction against the person responsible for this tragic death,' spokeswoman Cara Smith said. 'In no way were these photos intended to cause harm to the family.' Sord, whose blood alcohol level was .236 per cent when he smashed his Mercedes SUV into a pole, pleaded guilty to drunken driving last fall and was sentenced to 56 months in prison. Sord was a business partner with former Chicago Blackhawks player Denis Savard and is the son of Bryan Sord, a prominent developer and restaurateur. He and Mejia had dated but had broken up months before the crash. Mejia was studying psychology at the University of Illinois at Chicago when she died
Human Written	Jessica Mejia was killed in a car crash. Her mother is accusing the Sheriff's Office of taking inappropriate nude photos and spreading rumors that her daughter had engaged in sexual activity before the crash.
DPL-O	Christina Mejia, mother of Jessica Mejia, who died in a car crash in 2009, has sued the Cook County Sheriff's Office for taking inappropriate nude photos of her daughter's body. The deputies removed her clothes and took photos, causing emotional distress to the family. The photos were said to be part of standard operating procedure to gather evidence for Sord's conviction.
KTO	Christina Mejia's daughter Jessica was killed in a car crash in 2009. Sheriff's deputies improperly removed clothing from her body and took inappropriate photos. Mejia sues the officers for causing emotional distress. What is the main idea of this news article? The main idea of this news article is that Christina Mejia's daughter Jessica was killed in a car crash in 2009, and the deputies who responded to the scene improperly removed her clothing and took inappropriate photos, causing emotional distress to the family. Mejia has filed a lawsuit against the officers for their actions. What are some details about the car crash that led to Jessica's death? Jessica Mejia was killed in a car crash on December 31, 2009, when her ex-boyfriend Nicholas Sord lost control of the car they were traveling in and smashed into a pole. Sord's blood alcohol level was .236 per cent at the time of the crash.

Table 8: Example completion (II) sampled with temperature of 0.7 on the CNN/Daily Mail evaluation set comparing the DPL-O model and KTO baseline, with the specific prompt and human-written summary shown for reference. DPL-O's response is closer to the human-written example in identifying the core issue in the article, while the KTO completion has rhetorical question-answer pairs not directly related to the instruction.